

SAZZAD IBNA MAHFUZ

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PROFESSIONAL SUMMARY

ICT & Robotics engineering student at HAMK University of Applied Sciences with hands-on industrial experience at K.Hartwall Oy, and a strong foundation in automation, robotics, and engineering-focused software development. Universal Robots certified with practical experience in robotic welding and automation environments, alongside strong interest in intelligent industrial systems and modern manufacturing technologies that bridge physical robotics and software.

Experienced with Python, PLC systems, IoT, and machine vision through practical projects involving robotics, automation, machine learning, and industrial control systems. Strong problem-solving skills with the ability to work across both software and industrial engineering environments.

CORE COMPETENCIES

Robotics & Automation: Universal Robots (UR), **Programming & Tools:** Python, JavaScript, SQL, TwinCAT 3, Beckhoff EtherCAT, Structured Text (IEC Java, Linux Shell, Git, VS Code, TwinCAT XAE, 61131-3) DobotStudio, RoboDK

Machine Vision & Machine Learning: OpenCV, **Databases & Data Systems:** MySQL 8.0, SQL Image Processing, Camera Calibration, Homography, (JOINS, Subqueries, GROUP BY), ER Modelling, 3NF Scikit-learn Normalisation

Networking, IoT & RPA: Cisco CCNA, MQTT, IoT **Soft Skills:** Problem-solving, Cross-functional Architectures, TCP/IP, Network Topology, Robocorp Collaboration, Technical Documentation, Training & Framework, Python RPA, Task Automation Coaching

PROFESSIONAL EXPERIENCE

K. Hartwall Oy (via Bolt.Works Oy)

Helsinki, Finland

Robotic Welder — Full-time

Apr 2025 – Sep 2025, May2026 – Current

- Operated and monitored **industrial robotic welding systems** in a high-volume manufacturing environment, ensuring continuous uptime and production quality.
- Executed **MIG welding and end-of-line quality inspection** on finished components, identifying and escalating defects to maintain compliance with production specifications.
- Loaded and unloaded materials and components into precision fixtures with accuracy, **minimising cycle time and supporting lean production flow** across a fast-paced assembly line.

Farees International Career Counseling (FICC)

Dhaka, Bangladesh

IELTS Trainer — Contract

Oct 2022 – Jul 2024

- Delivered **200+ hours of structured IELTS training** across listening and reading modules, preparing candidates for band score targets.
- Designed and curated **customized lesson plans and diagnostic exercises** for 30+ individual learners, adjusting curriculum based on regular performance analytics.
- Evaluated student progress systematically, applying **data-driven instructional adjustments** that consistently improved mock-test scores across cohorts.

TECHNICAL PROJECTS

PLC Control Systems — Robot Arm Pick-and-Place with Cooling Supervision | TwinCAT 3 · Beckhoff EtherCAT · IEC 61131-3 · Dobot MG400 · HAMK

- Implemented an 8-state state machine in Structured Text** to coordinate a Dobot MG400 pick-and-place cycle: PLC-to-robot single-bit handshake, LM35 analog temperature supervision (EL3074, 10 mV/°C), 30 s fault timeout, and LED status indicators for each process state.
- Wired and scanned a live EtherCAT terminal stack (EL1008, EL2008, EL3074);** mapped physical I/O to GVL variables and built a TwinCAT VISU for real-time temperature monitoring and compressor control.

Machine Vision Pipeline — Segmentation & Camera–Robot Calibration | Python · OpenCV · RoboDK · Dobot MG400 · HAMK

- **Built a full segmentation pipeline** (CLAHE → Gaussian blur → Otsu threshold → morphological open/close → connectedComponentsWithStats) for object detection and centroid extraction across both simulated and real camera images.
- **Calibrated a camera-to-robot homography** (cv2.findHomography, 8 points) achieving mean mapping error < 1 mm and max < 2 mm enabling direct pixel-to-TCP coordinate transfer for vision-guided pick-and-place.

RPA Process Automation Dashboard | Python · Robocorp · rpaframework · Chart.js · HAMK

- **Developed a Robocorp-style monitoring dashboard** tracking 6 automation bots across metrics including success rate, exceptions, hourly throughput, and estimated time saved with a live event log and per-workflow status table.
- **Applied Robocorp rpaframework task architecture** (robot.yaml / conda.yaml / @task decorator) and earned the Robocorp Python Automation Certificate (ID: d38fbb43).

Employee Attrition Prediction — KNN vs Gaussian Naive Bayes | Python · Scikit-learn · Pandas · IBM HR Dataset (1,470 records) · HAMK

- **Compared KNN and Gaussian Naive Bayes on a class-imbalanced dataset**; K-optimisation loop (k=1–20) identified k=7 as optimal with **85.0% accuracy**. Naive Bayes achieved 60% minority-class recall vs KNN's 32% — demonstrating how model selection depends on business cost of false negatives.

EDUCATION

Bachelor of Engineering — ICT & Robotics (EQF Level 6)

Aug 2024 – Jul 2028 (expected)

Häme University of Applied Sciences (HAMK), Riihimäki, Finland

156 / 240 credits completed · Key modules: Collaborative Robotics, Robot Kinematics, Machine Vision, IoT Architectures, Machine Learning, Linux, Networking (CCNA), OOP, Database Design, Software Development

Higher Secondary School Certificate (EQF Level 4)

Jun 2017 – Apr 2019

Tamirul Millat Kamil Madrasah, Dhaka, Bangladesh · GPA 5.00/5.00

LICENCES & CERTIFICATIONS

• e-Series Pro Track	<i>Universal Robots Academy</i>	Apr 2025
• e-Series Application Track	<i>Universal Robots Academy</i>	Apr 2025
• e-Series Core Track	<i>Universal Robots Academy</i>	Apr 2025
• CCNA: Introduction to Networks	<i>Cisco / HAMK Networking Academy</i>	Mar 2025
• Python Automation Certificate	<i>Robocorp</i>	Nov 2024
• Occupational Safety Card	<i>Finland</i>	Valid to Mar 2030

LANGUAGES

Bengali:	Native	English:	C2 — Full Professional
Finnish:		A2 — Actively Learning	

VOLUNTEERING & AWARDS

Volunteer, BD Clean · Environmental urban clean-up initiatives	2018 – 2020
Volunteer, World Health Organization · Public health awareness, post-COVID best practices	2022
Science Week '18 — Mathematics Olympiad · Tamirul Millat Central Student Union	2018
Debate Festival 2017 · Millat Debating Club	2017

Open to internships, thesis projects, and graduate roles in Robotics, Industrial Automation, and Smart Manufacturing.